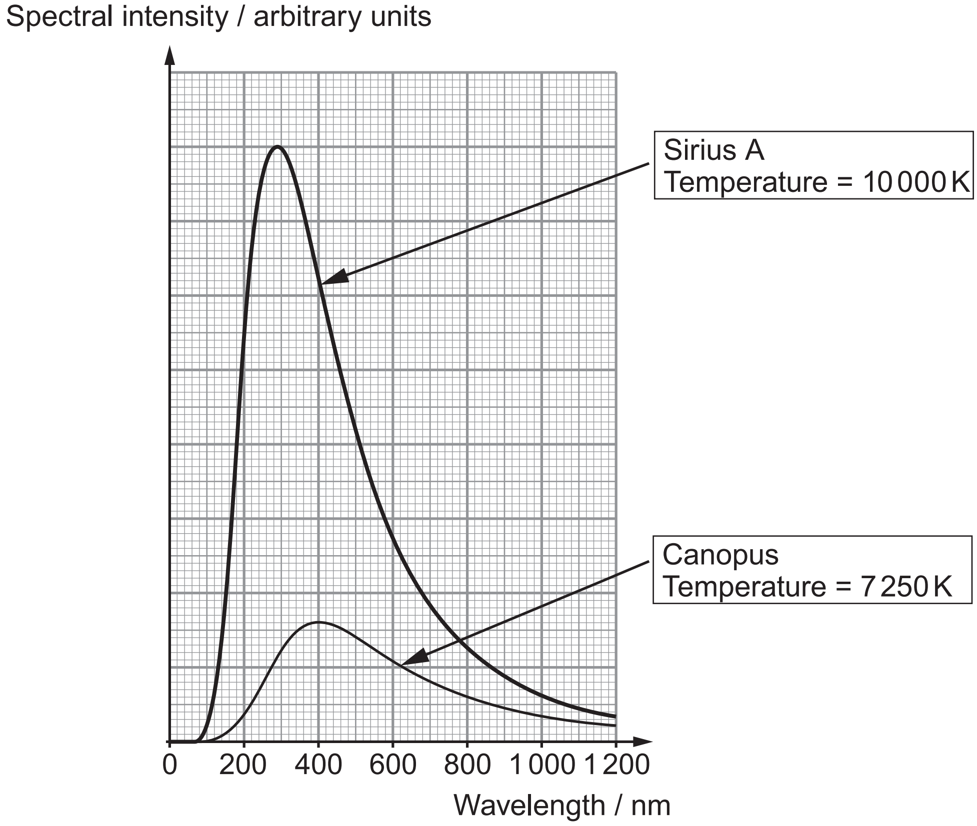
Examiner only

3. *(a)* Two of the brightest stars in the night sky are Sirius A and Canopus. The graph shows the continuous black body spectra for these two stars.



(i) Confirm that Wien's displacement law is valid for the stars. [3]

....

....

....

....

....

....

Examiner only

(ii) Astronomers state that one of the stars appears 'bluer' than the other. Explain how the spectra support this statement and state which star would appear bluer. [2]

....

....

....

....

(b) (i) The radius of Canopus is 4.97×10^{10} m and that of Sirius A is 1.19×10^{9} m. Show that the luminosity of Canopus is approximately 500 times the luminosity of Sirius A. [4]

....

....

....

....

....

....

(ii) Calculate the intensity of the radiation reaching the surface of the Earth from Sirius A. (Distance between Sirius A and Earth = 8.15×10^{16} m.) [2]

....

....

....

....

(iii) The intensity of the radiation reaching the Earth's surface from Canopus is **less than** that from Sirius A, even though Canopus has a **greater** luminosity than Sirius A. Explain this apparent contradiction. [2]

....

....

....

2420U101 07

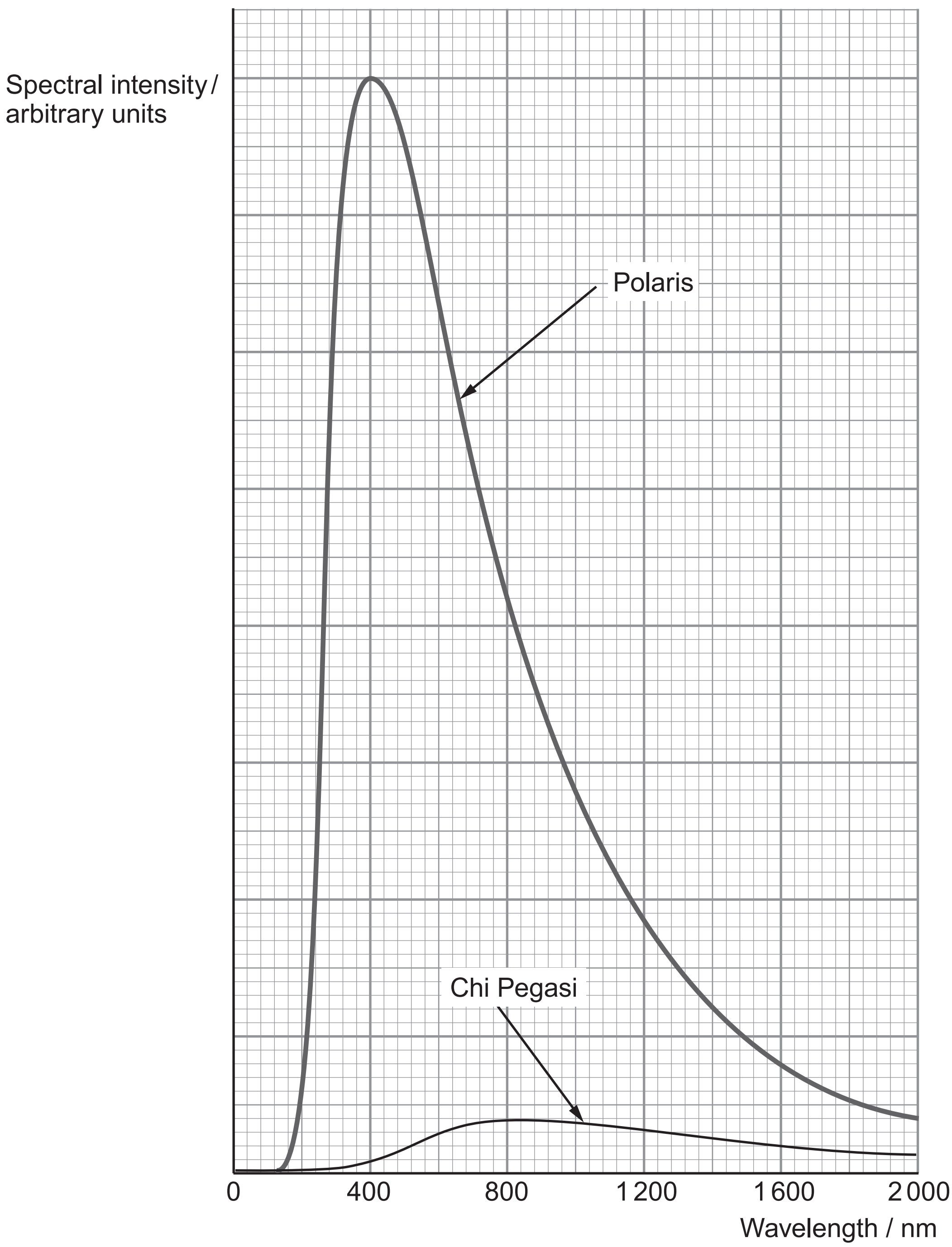
Examiner only

4. *(a)* Stars are very good approximations to *black bodies*. State what is meant by a *black body.* [1]

....

....

(b) The graph shows the black body radiation curves for the two stars Polaris (sometimes called the North Star) and Chi Pegasi (a red supergiant in the constellation Pegasus). The stars are equidistant from the Earth.



Examiner only

(i) Use the graph to state **three** differences between Polaris and Chi Pegasi. [3]

....

....

....

....

....

(ii) The surface temperature of Polaris is 7 250 K. How does the graph confirm this? [2]

....

....

....

(iii) Polaris is 431 light years from Earth and the intensity of radiation received on Earth from it is 4.05×10^{-9} W m^{-2}. Show that the luminosity of Polaris is approximately 8.5×10^{29} W. [1 light year = 9.46×10^{15} m] [2]

....

....

....

....

(iv) Calculate the radius of Polaris. [3]

....

....

....

....

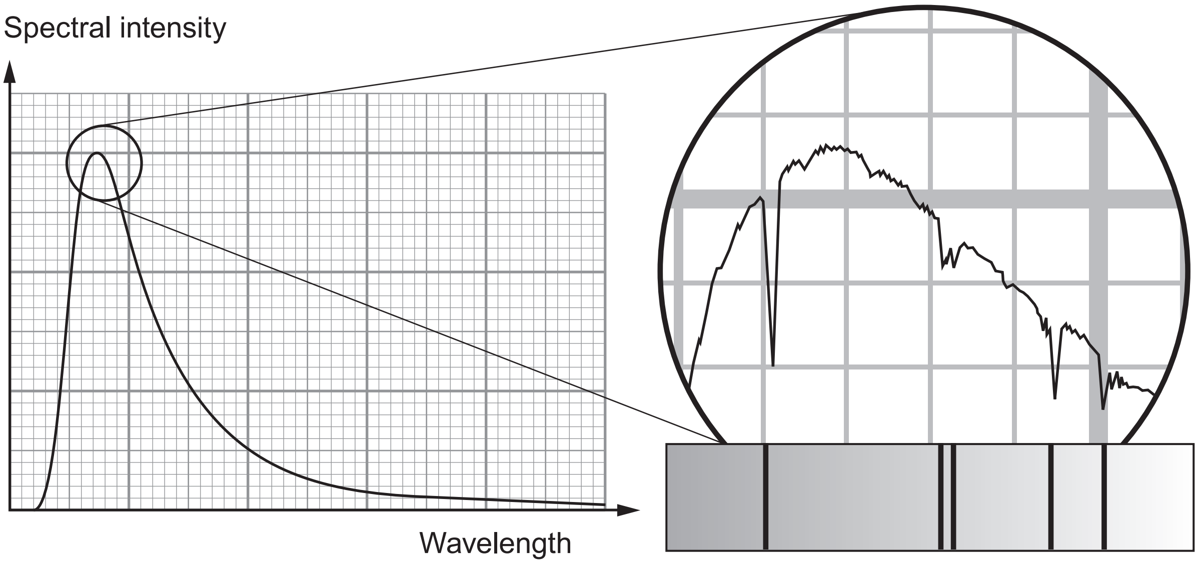
....

....

11

Examiner only

6. *(a)* A blackbody graph of spectral intensity against wavelength for a star is shown. A magnified section, showing the finer detail of the spectrum is also given. An associated line spectrum is also shown.



Explain how the graph and the spectra can be used to provide information about the star and the elements from which it is made. [6 QER]

....

....

....

....

....

....

....

....

....

....

....

....

....

....

....

Examiner only

(b) (i) Altair is the brightest star in the Aquila constellation. It is 1.58×10^{17} m away, and the intensity of its electromagnetic radiation reaching the Earth is $1.32 \times 10^{-8}\,\mathrm{W\,m^{-2}}$. Show that its luminosity is approximately 4×10^{27} W. [3]

....

....

....

....

....

....

(ii) Calculate Altair's **diameter** given that its surface temperature is 7 700 K. [3]

....

....

....

....

....

....

12

TURN OVER FOR THE LAST QUESTION

Examiner only

7. *(a)* Describe the main features of the spectrum of a star and state where in the star they arise. [2]

(b) The table gives some information about two stars.

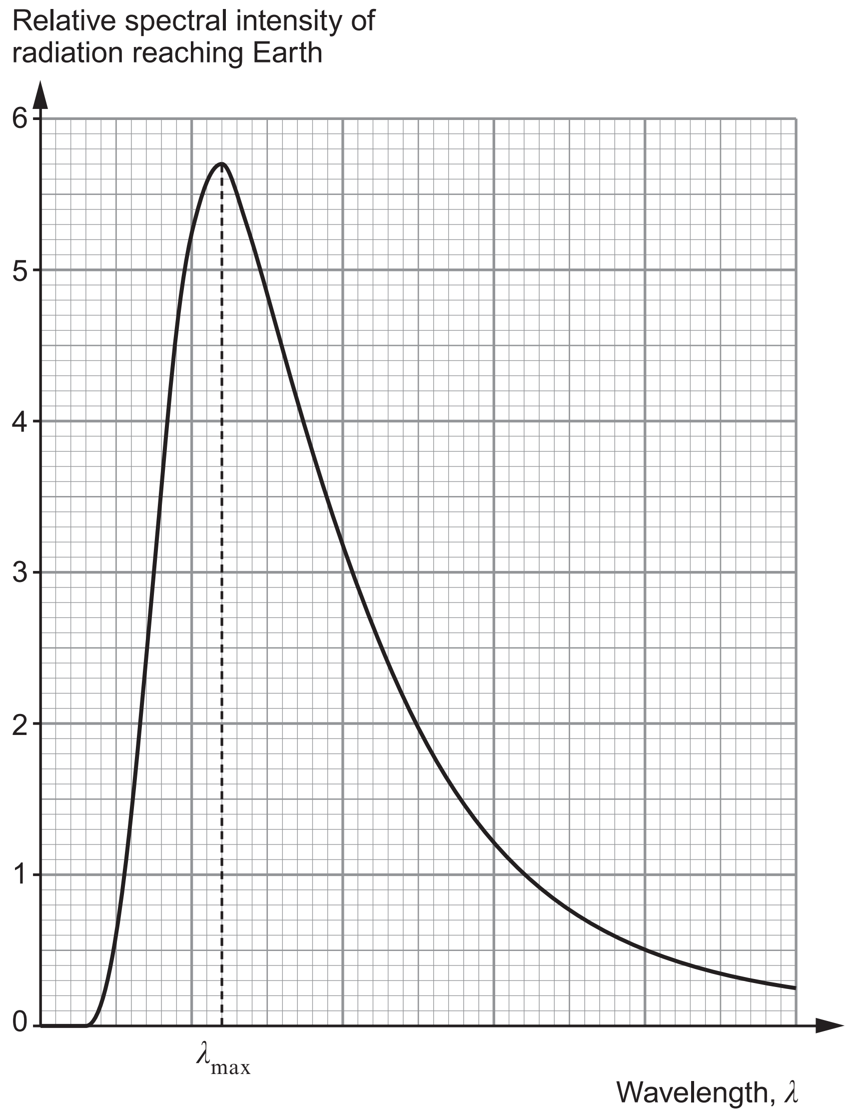
Star	Luminosity / W	Distance from Earth / m
Sirius	9.7×10^{27}	8.1×10^{16}
Vega	1.5×10^{28}	2.4×10^{17}

(i) Determine the ratio:

$$\frac{\text{Intensity of radiation reaching Earth from Sirius}}{\text{Intensity of radiation reaching Earth from Vega}}$$ [3]

Examiner only

(ii) The two stars have similar surface temperatures. Below is a black body radiation curve for Sirius. Using your answer to *(b)*(i), sketch the expected black body curve for Vega on the same axes. [2]



(iii) Determine λ_{max}, given that the surface area of Sirius is $1.8 \times 10^{19}\,m^2$. [4]

....

....

....

....

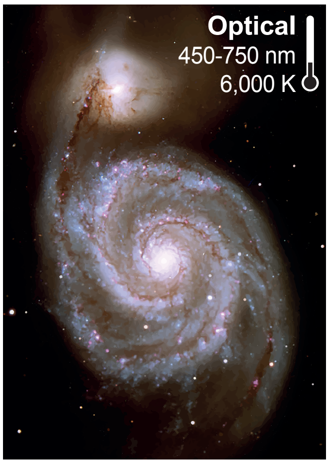
....

....

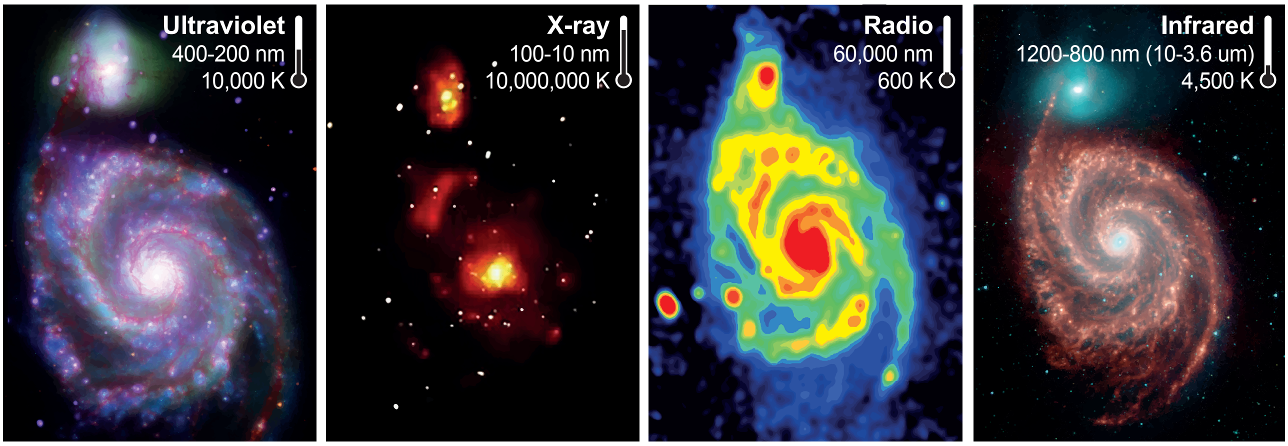
....

Examiner only

(c) The image below is of the whirlpool galaxy, M51 (or NGC 5194). This is one of the first galaxies to be photographed by astronomers.



Subsequent images of the **same galaxy** are shown below.



Describe how these developments in observational astronomy have advanced the study of the whirlpool galaxy. [3]

....

....

....

....

....

....

14

END OF PAPER

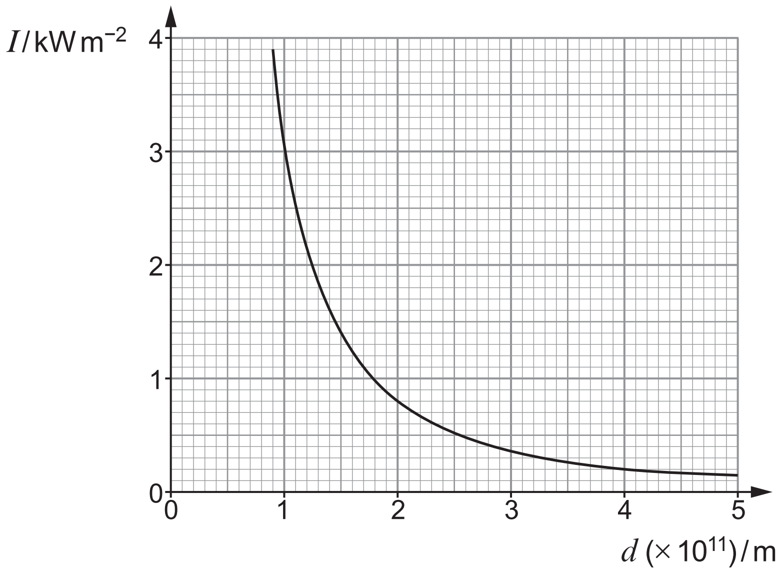
Examiner only

7. (a) Stars are very good approximations of *black bodies*. Explain what is meant by the term *black body*. [1]

....

....

(b) The graph shows how the intensity of electromagnetic radiation, I, from the Sun varies with the distance from its centre, d.



(i) Confirm that the graph shows an inverse square relationship between I and d. [2]

....

....

....

....

....

(ii) Show that the Sun's luminosity is approximately 4×10^{26} W. [2]

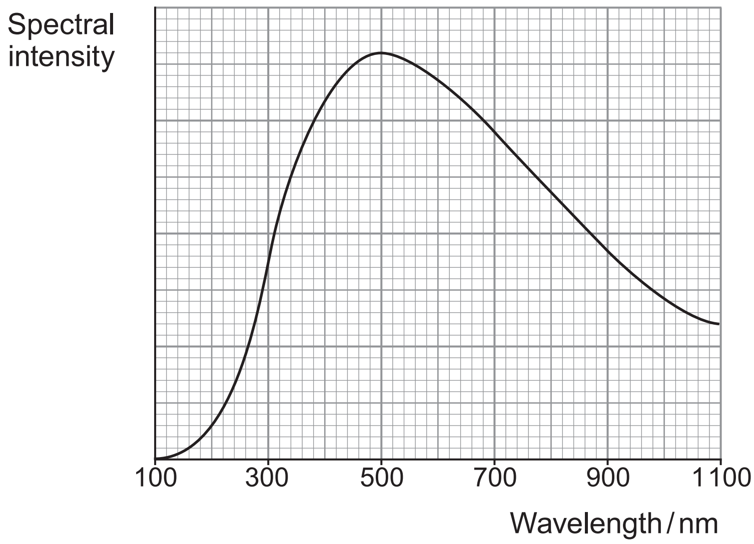
....

....

....

Examiner only

(c) The graph shows how the intensity of the radiation incident on the Earth from the Sun is distributed across the spectrum.



Use information from this graph, along with your answer to (b)(ii), to calculate the radius of the Sun. [4]

....

....

....

....

....

....

9

END OF PAPER

Examiner only

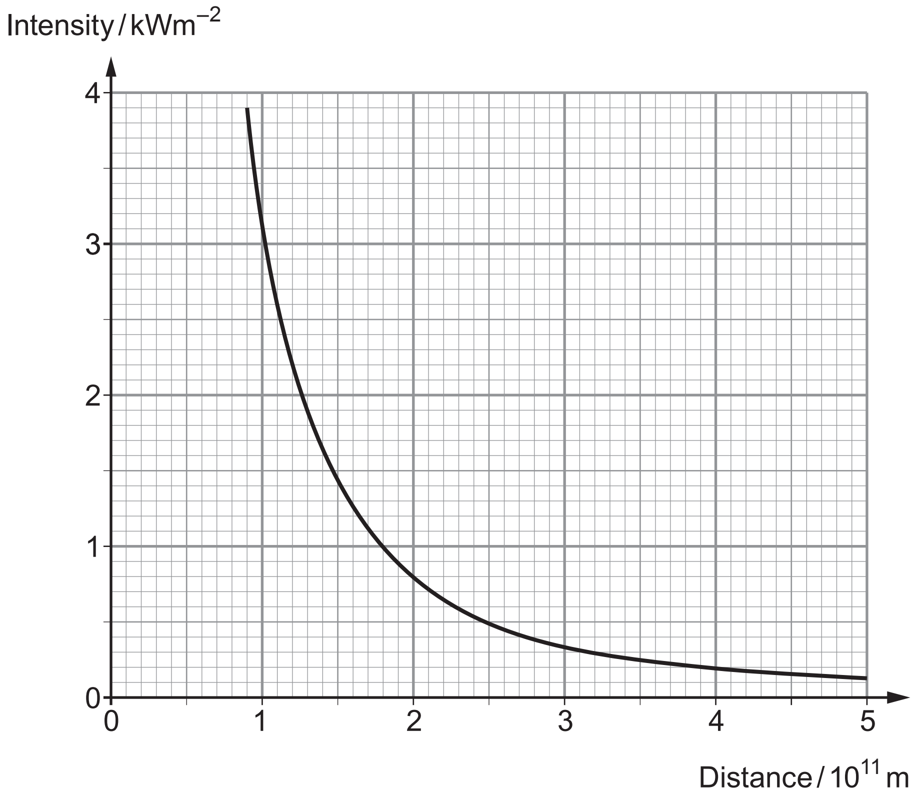
8. (a) The solar spectrum includes an absorption spectrum. Describe the appearance of this spectrum **and** state where in the Sun the absorption occurs. [2]

....

....

....

(b) The graph shows how, outside the Sun, the intensity of electromagnetic radiation from the Sun varies with the distance from its centre.



Use information from the graph to determine the total power emitted by the Sun. [3]

....

....

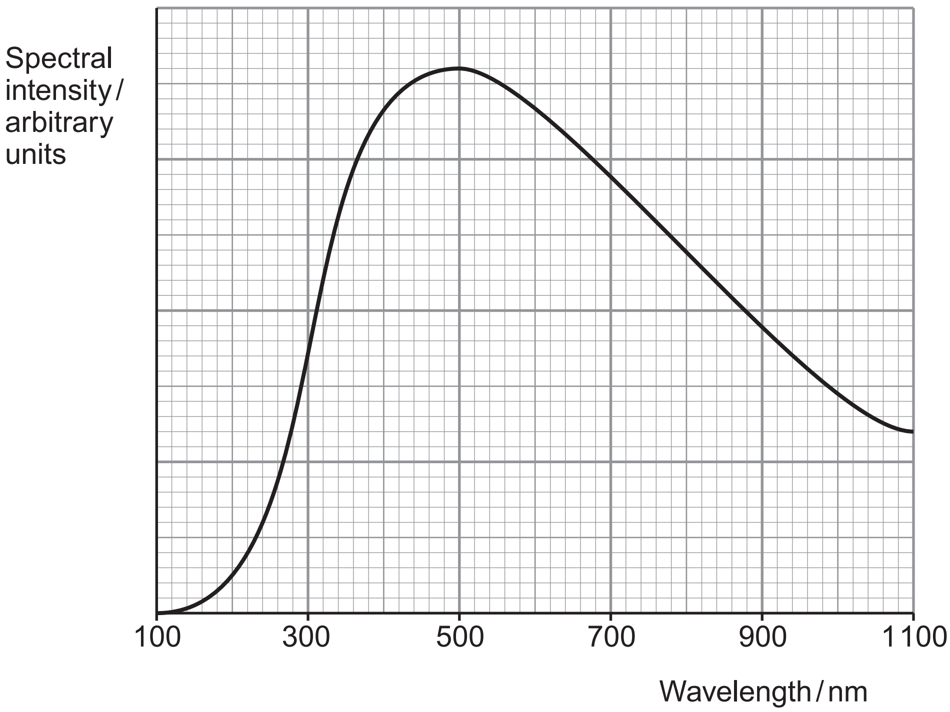
....

....

....

Examiner only

(c) The diagram below shows how the intensity of the radiation incident on the Earth from the Sun is distributed across the spectrum.



Determine whether or not the answer you obtained in part (b) and information which can be obtained from the above spectrum are consistent with each other.
[Surface area of the Sun = $6.2 \times 10^{18}\,m^2$] [4]

....

....

....

....

....

9

END OF PAPER

Examiner only

7. (a) Explain what is meant by the term 'multiwavelength astronomy'. [2]

....

....

....

(b) The Stefan constant, σ, has the unit $\mathrm{W\,m^{-2}\,K^{-4}}$. Express this in terms of base SI units. [2]

....

....

....

(c) The wavelength of the peak emission of the bright star Arcturus is measured to be 674 nm.

(i) Calculate the star's temperature. [2]

....

....

....

(ii) The visible spectrum extends from approximately 400 nm to 700 nm. State the colour of Arcturus. [1]

....

Examiner only

(d) A website claims that Arcturus is about 37 light years from the Sun. Use the following information and your answer to (c)(i) to justify this claim. [5]

Diameter of Arcturus = 2.78×10^{10} m
Intensity of radiation received on Earth from Arcturus = 3.09×10^{-8} W m^{-2}
1 light year = 9.46×10^{15} m

....

....

....

....

....

....

....

....

....

12

END OF PAPER